

Hadi Fathipour

Backend & AI Engineer

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Professional Summary

Backend & AI engineer with experience building scalable backend systems using Django, FastAPI, and asynchronous architectures. Strong background in designing RESTful APIs, role-based access control systems, containerized services, and real-time data processing pipelines. Experienced in translating complex domain requirements into reliable, production-grade backend services.

Technical Expertise

- **Programming:** Python (NumPy, Pandas), SQL, C++, MATLAB, HTML/CSS
- **Backend & DevOps:** Django, Django REST Framework, FastAPI, AsyncIO, Docker, Docker-py, Celery, Kafka, RabbitMQ, Linux, Git, REST, WebSockets, gRPC
- **ML & AI:** TensorFlow, Keras, scikit-learn, Transformers, BERT, XGBoost, SVM, RNN, ARIMA
- **Data & DB:** PostgreSQL, MySQL, Redis, SQLAlchemy, SQLModel, Django ORM

Experience

Backend & AI Developer — Sanam.pro (RobTK)

Moscow, Russia

2025 – 2026

- Designed and implemented backend services using FastAPI and Django for real-time drone command and telemetry management.
- Built secure role-based authentication systems and administrative control panels using Django ORM and permission frameworks.
- Developed modular service architecture separating API, business logic, and data layers.
- Implemented path-planning algorithms with NetworkX and optimized geo-computation pipelines using Shapely, GeoPandas, GeoPy, and Cython.
- Integrated MAVSDK and PyMAVLink for reliable real-time communication.

Freelance ML & Backend Engineer — Self-employed

Remote

2021 – Present

- Delivered end-to-end backend and ML solutions across NLP, analytics automation, and time-series systems.
- Designed RESTful APIs using Django and FastAPI for scalable business workflows.
- Deployed containerized backend services using Docker in production environments.

Algorithmic Trader & Systems Developer — Self-employed

Remote

2022 – Present

- Designed and deployed algorithmic trading systems and backend data pipelines for cryptocurrency markets.
- Built containerized execution services for reliable unattended server operation.

- Applied forecasting models (Transformers, RNNs, ARIMA) to evaluate strategy performance.

Key Projects

Urban Drone Backend & AI Control System

Tools: FastAPI, Django, AsyncIO, NetworkX, GeoPandas, Shapely, MAVSDK

- Developed backend systems to manage autonomous drone behavior and enforce restricted-zone compliance.
- Built asynchronous services and real-time monitoring dashboards.
- Implemented structured database models and permission systems using Django ORM.

IFUND: Prop-Trading Platform Backend & Infrastructure

Tools: Django, FastAPI, Docker, Docker-py

- Developed backend services to manage trading challenge provisioning and user account isolation.
- Designed Django-based administrative dashboards and data models.
- Implemented container lifecycle management via REST endpoints.

TONPAY: Telegram Crypto Wallet

Tools: Django ORM, SQLAlchemy, AsyncIO

- Designed an asynchronous cryptocurrency wallet backend integrated with Telegram.
- Built scalable database schemas supporting multiple wallets per user.

Pycoin: Algorithmic Trading Framework

Tools: Python, Binance API

- Developed an open-source framework for market data retrieval and technical analysis.
- Implemented modular tools for trend detection and probabilistic strategy modeling.

Education

KNTU University of Technology — B.Eng. (GPA: 3.81)

2017 – 2022

KNTU University of Technology — M.Sc. Computer Engineering (AI)

2026 – Present

Hobbies

Mountain climbing, gym, swimming, reading, and film.